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Validity of a Simulation Game as a Method for History Teaching

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Abstract

The aim of this research is, first, to determine the validity of a simulation game as a method of teaching and an instrument for the development of reasoning and, second, to study the relationship between learning and students' behavior toward games. The participants were college students in a History of International Relations course, with two groups participating in a simulation game and one serving as a control group. The results show that the experimental groups had higher scores on a test of comprehension and that participants within these two groups who were identified as operating at a level of formal reasoning (as evaluated by an arrangement test) obtained the highest knowledge scores. Learning style was found to be an important variable in explaining motivation toward the game.

Keywords

behavior, efficiency of games, experiments on games, history, learning style, reasoning, validity

Research on the development of formal operations—particularly in Québec, Canada (Torkia-Lagacé, 1980)—has shown that teaching methods rely on inadequate assumptions about theoretical learning. For many students, conceptual development drags through the secondary and college levels (students aged 12 to 19 years).

The research conducted by Torkia-Lagacé (1980) shows that when compared to students of health and physical sciences, a smaller proportion of humanities and social sciences students have reached a high level of formal operations. This state of affairs may be explained in several different ways: First, teaching is inadequate; second, formal operations develop later in the field of humanities and social sciences (horizontal decalage); third, the research carried out so far has been inadequate or biased.

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Up to now, research on cognitive development, particularly within the epistemological framework developed by Jean Piaget, has concentrated on logico-mathematical operations. Tests to measure formal intelligence skills were principally developed for those operations. It is therefore difficult to use the discoveries made with logico-mathematical thought to apply one of the above explanations to the situation in the humanities and social sciences.

Hallam (1969a) tried to apply the pedagogical theories of Jean Piaget to the teaching of history. In a preliminary experiment, he was able to test a series of criteria that evaluated students' reasoning in analyzing historical documents. His conclusion was that adolescents reach formal levels of reasoning much later in history than in other fields. As a result, he recommended that major changes be made in the customary methods of teaching history (Hallam, 1969b). According to Hallam, the teaching of history should strive to eliminate preoperatory thinking and so move students up to formal reasoning by the end of secondary school. Hallam's recommendations were as follows:

- Avoid confusion by reducing the number of conflicting variables in the problems studied.
- Encourage interaction between the students to compare values and judgments.
- Use visual support within methods based on personal discovery.

We can summarize Hallam's conclusions by saying that he favors all methods that make the past come alive. Although Hallam did not specifically mention simulation games, as a teaching method they answer his criteria very well. Based on student interaction, the simulation game creates simplified problems and makes them come alive. It offers a likely prospect as a method to teach history.

The concept of the simulation game is a multiple one, involving different operational definitions. Schild (as cited in Lecavalier, 1971) uses this one:

> The simulation game, as opposed to the role-playing game, does not try to provoke certain feelings in the participant, but rather places him in a context, with certain givens, where he must attempt to increase his chances of winning and decrease his chances of losing. Givens and results are more explicit in a simulation game than in a role-playing game.

Cavanagh (1975) gave several reasons favoring the use of simulations and role-playing games in the teaching of history. These reasons are closely related to Hallam's principles. The major ones are as follows:

- The simulation game takes the student beyond any determinist approaches and allows him or her to understand the uncertainties of the past.
- Lectures must function sequentially, although historical manifestations are simultaneous and simulation games can correct this.
- The simulation game introduces other versions and shadings usually underrepresented in historical studies, such as the loser's point of view.

- The simulation game encourages self-learning and the study of historical processes as well as data.

Cavanagh (1975) rejected the argument that simulation games deform reality. Fidelity to an external model is not necessarily a criterion, because, as Shirts pointed out (as cited in Cavanagh, 1975), the object may very well be to exaggerate, modify, or simplify situations to better study them.

The research carried out so far has not succeeded in clearly showing the advantages of simulation games for teaching. This can be explained by several factors. To begin with, a simulation game can be operationalized in many ways. Also, simulation games often pursue goals different from those of more customary methods, making evaluation and comparison a difficult and uncertain matter. Finally, many experiences with simulation games have been timid, and it is hardly realistic to expect a major impact after a few hours' use.

In a review of the problem regarding the efficiency of simulation games as a method of teaching, Foster et al. (1980) concluded that the method has proven its superiority over more customary methods. The authors suggest that the evaluation of simulation games must take into account subjective understanding, or *Verstehen*, which is a process of understanding by way of personal experience. Students understand events better as actors than as mere observers. Foster et al. discovered that the simulation game considerably increased the students' subjective understanding. Although simulation games and role-playing games may not be the best methods of transferring information, they do encourage new types of learning and learning integration.

Because the efficiency of the simulation game as a pedagogical device depends on many factors, it is difficult to compare different studies: Bredemeier and Greenblat (1981) identified a number of variables that affect outcome and so make the comparison of simulation games and customary methods problematic:

Set effect: Set effect posits that the presentation of the game and the students' expectations will affect learning.

Experience: Experience includes the teacher's personality, variations in procedure, and the type of debriefing.

Personal variables: Personal variables include group dynamics, personality differences, and cognitive styles—that last of which, the authors suggested, has been underexplored.

Game-playing skills: Seginer (1980) suggested that game-playing skills are quite different from academic skills, which depend mostly on verbal ability and abstract thinking.

Few studies have controlled all these factors, and few discoveries have been confirmed. The true efficiency of the simulation game has not yet been determined, for two reasons. On one hand, most research has compared simulation games with customary methods, without attempting to match personality, types of learning, and

Table 1. The Simulation Game and Piaget's Theory of Cognitive Development

Assimilation (Play or Game Side)	Accommodation (Imitation or Simulation Side)	Student's Conception at Each Stage
Using the rules: Assimilation	*Learning the context of the game:* The map, the resources' management, each scenario's constraints	*Play:* Game is a symbolic representation through the material.
Coordination of rules: Reciprocal assimilation of one rule by another	*Learning strategies:* Rules are grouped and classified; rules are differentiated and specified.	*Game of rules:* The game is one form of representation, and history is another. Events of the game and events of history are seriated separately.
Coordination of strategies: Repetitive assimilation (using strategies that are successful); generalized assimilation (conditions when strategies are effective)	*Learning outcomes:* Grouping strategies as a function of their outcomes: differentiating and specifying, identifying what makes them produce a specific outcome	*Game of rules about history:* Some outcomes of the game are similar to some outcomes in history. Events occurring in the game and events known to have occurred in the past are coordinated.
Coordination of outcomes: Of the games among themselves; of the game with historical events	*Learning from the game:* Grouping of causal factors, such as alliances, geographic proximities, the role of colonies, and so on	*Simulation game about history:* Some hypotheses link causal factors of game events with historical factors that have had similar effects.
Coordination of the game's outcomes: Within a set of other games' possible outcomes	*Learning about the game, or meta-game:* Creating new rules and new game constraints; considering a game as one form of history and history as one form of game	

teaching methods. On the other, no research established how well the students learned the game; all such research took for granted that once the students were playing correctly, they were ready to learn. Our experience with simulation games has shown that students do not assimilate the game in identical fashions. This factor has a necessary impact on how they learn through the game.

For many students, the game is only a game until they have understood the symbolic import of the material. These students pay more attention to the rules than to the meaning of ongoing events. Just as Hallam (1969a, 1969b) did for the levels of development of historical reasoning, we have come to postulate stages in the assimilation and accommodation of a simulation game, as inspired by the theory of Jean Piaget. Table 1 illustrates the students' different operational levels within the framework of an international relations game, as well as the evolution of their vision of the game. This model was designed on the basis of a study of the students' papers (Laveault & Corbeil, 1983), and it shows that learning through the games is intertwined with learning the game itself; that is, the quality of the student's assimilation depends on his or her capacity to go beyond the mere coordination of strategies.

Our experience shows that the students move quickly through the first stages of learning the rules and the strategies. Students remain for a long time at the next level—the coordination of strategies—which allows them to acquire the competence necessary to rapidly learn from the game. From this point on, learning again accelerates, but now the student is learning from the game, not about the game.

We are dealing here with a variety of historicism: The historian is trying to put himself or herself in the historical characters' shoes, and the teacher is trying to help his students do the same but actively. History is a personal reconstruction in which we must remember that our sense of a long-lasting reality is a carefully protected illusion. In such a perspective, it becomes meaningful to create a game model to be used as a tool by the student. It is in this manner that the game on international relations was slowly developed, over several years, which is the subject of our research.

This research had three objectives. First, it intended to test the validity of simulation games as a teaching tool in a course on the history of international relations. Second, it tried to establish whether a simulation game could be used as a tool to further the development of operational thought in adolescents. The fields of operational development and simulation gaming have had little contact, yet Piaget's formal stage is precisely defined by the extension of reasoning from "real" objects to hypothetical conditions. This is exactly what is involved in the use of simulation games. The last objective was to discover possible links between learning and adolescent behavior, in terms of attitudes and learning styles.

The research isolated three control variables in an attempt to generalize findings beyond this particular simulation game:

The participant: Which students benefited the most as a result of participation in the game? All? Those at one or another operational level? Those with specific preferences for some learning styles?

The activity: The participant in a simulation is subject to rules and conditions. Do changes in conditions, through different scenarios, favor different learning objectives?

The material: To what extent do concrete supports, in the form of game materials, help one understand abstract concepts, and does the nature of the game material favor learning objectives?

Each variable relates to a hypothesis:

Hypothesis 1 (participant hypothesis): Concrete-level (or preformal) participants will benefit more from participation in the game. Formal participants will succeed more easily, but concrete ones will have a greater rate of progress.

Hypothesis 2 (activity hypothesis): Modifying scenarios, in terms of rules and conditions, will assist learning of specific objectives.

Hypothesis 3 (material hypothesis): Concrete supports are necessary to manipulation and so favor the discovery of important factors through the materialization of concepts in maps and pieces.

Method

Participants

Three CEGEP-level history classes were involved in the research (i.e., Collège d'Enseignement Général et Professionel [College of General and Vocational Education], Québec, Canada; students aged 17 to 25). A class in Trois-Rivières was used a control group, whereas two classes in Drummondville were the experimental groups using the simulation game.

Instruments

The research involved instruments to measure the dependent variables, as well as instruments to control the game as an independent variable. Six were used to measure the dependent variables:

1. the LAM-30P questionnaire, created by Claude Lamontagne (Lamontagne, Lamontagne, & Lamy, 1982), used to identify learning styles;
2. tests of operational development (Laveault, 1981; Noelting, 1982);
3. a questionnaire on attitudes, developed for our project, relating to the following conditions—the role of the game material, the activity, the impact of the teacher's character traits and the students' participation (Cuisinier & Laveault, as cited in Corbeil, Cuisinier, & Laveault, 1984);
4. a questionnaire on the students' opinions regarding knowledge in general and the history course in particular (Corbeil et al., 1984);
5. an objective test to measure students' knowledge level at the end of term (Corbeil et al., 1984); and
6. a five-series test of comprehension in historical analysis.

The last item is given in two phases. A document, or a series of documents, is first distributed with no contextual information. The students have to answer questions on this document. In the second phase, the students are asked to check their first answers when provided with new information. The objective here is to measure a student's capacity to use the context and to group new information to solve a historical problem (Corbeil et al., 1984).

The simulation game involves the movement of pieces on a world map—specifically, Antarctic polar projection—within each team's resources and in the pursuit of its objectives. The situation is loosely based on the DIPLOMACY game (originally distributed by Waddington; the only current publisher is Avalon Hill [owned by Hasbro]).

Procedure

The developmental and learning tests were administered to all three groups at different times during the term. Only the experimental groups answered the questionnaires on

Table 2. Research Schedule

Period	Control Group	Experimental Group
Beginning of term	Tests on operational development	Tests of operational development Questionnaire on learning style
During the term	Regular lectures	Simulation game. Test of comprehension in historical analysis after each scenario (Tests 1–5): Group l, Tests 1–5; Group 2, Tests 5–1
End of term	Objective test: historical data Test of comprehension in historical analysis (Tests 1 and 5) Test of opinions	Objective test: historical data Tests of attitudes and opinions

attitude and learning style, given that their purpose was to determine the character types for which the method was most effective. All three groups answered the opinion questionnaire.

The two experimental groups participated in the simulation game in four scenarios of increasing order of difficulty. The control group attended its ordinary lectures. The students in the experimental groups each received a collection of historical documents, including a chronology. Table 2 shows the research schedule.

Results

1. The Game and Learning

The test results for knowledge and comprehension of both experimental groups were compared to those of the control group. Table 2.1 shows that no significant difference occurs on the knowledge test between the experimental groups (max = 32.00; Group 1, M = 17.42; Group 2, M = 15.65) and the control group (Group 0, M = 16.86). Group effect is not significant, F = 0.912, p = .410.

Table 2.2 shows that the experimental groups had significantly better results on the first comprehension test (max = 5.00; Group 1, M = 2.67; Group 2, M = 3.24) than the control group did (Group 0, M = 1.73), F = 5.578, p = .007. Results on the second test favored the experimental groups but not significantly so (Group 1, M = 2.42; Group 2, M = 2.29; Group 0, M = 1.87).

The formal level of the students is a good predictor for only the test of historical knowledge, r = .53, p = .049, n = 21. Table 2.3 shows the results of a statistical analysis on each student's score on this test. A significant difference of 4 points occurs between the scores of the formal and preformal students on this test. However, no statistically significant difference was found between formal and preformal students for the comprehension tests, r = .23, p = .614, n = 21.

Table 2.1. Test of Historical Knowledge

Source	*df*	*MS*	*F*	*p*
Groups	2	12.092	0.912	.410
Error	40	13.263		
Total	42	13.207		

Note: Experimental groups versus control group ($n = 31$).

Table 2.2. Historical Comprehension: Test 1

Source	*df*	*MS*	*F*	*p*
Groups	2	9.068	5.578	.007
Error	41	1.626		
Total	43	1.972		

Note: Experimental groups versus control group ($n = 44$).

Table 2.3. Historical Knowledge: Formal Students Versus Preformal Students (Experimental Groups Only)

Source	*df*	*MS*	*F*	*p*
Intergroup	1	108.52	8.84	.006
Intragroup	31	12.27		
Group	*n*	*M*	*SD*	
Preformal students	19	15.47	3.37	
Formal students	14	19.41	3.80	

2. Attitudes

Seven questions of the questionnaire on attitudes were singled out for a preliminary analysis of the relationship between attitude and test results: absenteeism, involvement, interest for the material, motivation compared to a lecture course, diversity of possible activities, reaction to teaching by simulation game. The only significant relationship was between involvement and historical comprehension test results, $r = .50$, $p = .011$.

3. Declared Attitudes and Learning Styles

The reaction of the students to the simulation game can be summarized as follows: 64.7% found the game more motivating and more diversified than a lecture course; 20.6%

Table 3. Attitude: Second Chance, Absenteeism, Motivation

Attitude: Learning Style	Correlation (Spearman Rho)	*p*
Second chance		
Proprioceptive cinematic	.5011	.003
Proprioceptive temporal	.4519	.007
Sensorimotor	.5686	.001
Kinesthetic	.4674	.005
Transactional	.4485	.007
Absenteeism		
Auditive	.4560	.006
Motivation		
Sensorimotor	.4297	.009

found it equally motivating and 23.4% found it equally diversified; 97.1% thought it required an equal or less amount of work than other courses (less work: 55.9%; equal work: 41.2%).

Certain questions asked the students directly about their appreciation of the course:

5. Would you take this course again, knowing what you now know? (second chance)
7. Were you absent from class?
9. Were you very involved in the game?
10. Did you find the material interesting?
14a. Is this course more or less motivating than a lecture course?
14b. Does this course offer more or less diversity than a lecture course?
20. Did you like the game method of teaching?

The most significant relationships between attitude and learning styles were as follows: Sensory-motor, kinesthetic, and proprioceptive students enjoyed being able to study abstract matter with tools appropriate to their styles. Transactional students reacted favorably to the possibility of controlling their environment and their learning activity. Auditive students tend to be absent most often. The more students tended to prefer sensory-motor learning, the more they found the course motivating (Spearman rho = .4297, p = .009; see Table 3).

4. Relationship Between Attitude and Operational Level

To what extent did students' operational levels (as determined by tests of combination and proportion) affect their attitudes in the course? This was determined by establishing the percentage of formal and preformal students who answered yes or no to the following two questions:

Table 4.1. Frequency of Formal and Preformal Students Who Would Take the Course Again (Second Chance): Experimental Groups Only

	Proportion (n = 34)		Combination (n = 32)	
Option	Formal n (%)	Preformal n (%)	Formal n (%)	Preformal n (%)
Take course again	21 (75.0)	7 (25.0)	14 (53.8)	12 (46.2)
Not take again	4 (66.7)	2 (33.3)	2 (33.3)	4 (66.7)
Total	25 (73.5)	9 (26.5)	16 (43.8)	16 (56.3)

Note: Proportion χ^2 = 0, significance level = 1; Combination χ^2 = 0.13, significance level = .91.

Table 4.2. Frequency of Formal and Preformal Students Who Thought That 51% or More of the Class Liked the Course: Experimental Groups Only

	Proportion (n = 34)		Combination (n = 32)	
Class Percentage	Formal n (%)	Preformal n (%)	Formal n (%)	Preformal n (%)
≥ 51%	21 (72.4)	8 (27.6)	11 (40.7)	16 (59.3)
< 51%	4 (80.0)	1 (20.0)	3 (60.0)	2 (40.0)
Total	25 (73.5)	9 (26.5)	14 (43.8)	18 (56.3)

1. Would you take the course knowing what you now know? (second chance)
2. Do you think more than 50% of the class liked the course?

Results show no significant difference between formal and preformal students on these questions, as shown by Tables 4.1 and 4.2.

5. Relationship Between Opinions, Attitudes, and Learning Styles

Results show no significant difference between formal and preformal students in their expressed opinions on the game, on knowledge, and on history. Differences do exist if we divide the students into (a) those who would take the course again and those who would not and (b) those who think that less than 50% liked the course and those who think that more than 50% liked the course.

A greater proportion of those who would take the course again agreed with the following statements (see Table 5):

Opinion A: Learning a game is as good an education as learning a science.
Opinion B: Understanding why we win a game helps us understand other ideas.

Table 5. Frequencies of Expressed Opinions According to Attitude: Experimental Groups Only (n = 34)

		n (%)	n (%)	
Opinion	Choice[a]	Would Not Take Course Again	Would Take Course Again	χ^2
A	1	4 (80.0)	5 (17.2)	9.20**
	2	1 (20.0)	8 (27.6)	
	3	0 (0.0)	16 (55.2)	
B	1	2 (40.0)	0 (0.0)	15.13**
	2	1 (20.0)	1 (3.4)	
	3	2 (40.0)	28 (96.6)	
		More Than 51% Liked the Course	Less Than 51% Liked the Course	
C	1	7 (25.0)	0 (0.0)	10.89**
	2	1 (3.6)	3 (50.0)	
	3	2 (71.4)	3 (50.0)	

a. Choices: 1 = *disagreement*, 2 = *no opinion*, 3 = *agreement*.
**$p < .01$.

Students who think that more than 50% liked the course believe:

Opinion C: The world is no better and no worse than it was fifty years ago.

Discussion

Our research has generated a large number of data on 65 college-level students; 44 of these in the experimental group were observed continuously over a 4-month term. In sum, 330 variables were identified involving personality, intellectual capacity, scholastic results, attitudes, and opinions. With so many variables, an intensive analysis was preferred to an extensive superficial one. A true statistical multivariate study would require 500 students and 20 game-experienced teachers, which was impossible in the present circumstances.

The unique character of the game under observation is that it requires research that is simultaneously fundamental and applied. To this end, we have attempted an intensive, almost-clinical approach that, when combined with usual statistical methods, has allowed us to generally confirm our hypotheses.

Our first objective was to test the simulation game as a tool for teaching history, in terms of both knowledge acquisition and comprehension. In terms of knowledge, the game is, on average, as efficient as a lecture course, which most likely means that it is more efficient with some learning styles and less with others. It seems surprising that the lecture is not more efficient at this level. The most likely explanation is that a lecture pressures students into memorizing a specific set of data; however, the students who were involved in the game had to choose, among a variety of historical data, those most useful their strategic situation. Quantitatively speaking, the results come out to be the same in terms of comprehension; that is, our results suggest greater efficiency for the game. In other words, the experimental groups had superior results. Although the difference is significant in only a single case, one has to consider that such difference was obtained in only 4 months.

The game resulted in equally good results for formal and preformal students. No significant difference appears in the results of the comprehension tests between the two groups. However, we see a difference in the knowledge control test, indicating that students at the formal level of the combination test were able to acquire more knowledge from the game.

The game does seem to motivate certain learning styles (e.g., sensory-motor). Operational level seems to make no difference.

The students definitely liked the course: A great majority would take it again and so thought that more than 51% of their peers enjoyed the simulation game.

Another objective of our research was to evaluate the efficiency of the game as a tool for developing operational thought in adolescents, as well as a tool for learning in general. Here we advanced three hypotheses.

Our first hypothesis presumed that preformal students would not succeed quite as well as formal students, but that they would make more gains. On one hand, the five historical comprehension tests show no significant difference between formal and preformal students, which suggests that participation in the game helped improve understanding in the preformal students. On the other, formal students, as demonstrated by the combination test, retain more information, most likely because they are better at organizing it. The first result is particularly meaningful because it is rare to find no difference between formal and preformal individuals in Piagetian research. In terms of teaching, this means that the simulation game was of most benefit to those students who most needed help at the level of concrete operational thought.

Our second hypothesis suggested that changes in the scenarios, in terms of rules and factors, would aid in attaining learning objectives. Our results show that the course objectives were attained, but it is impossible at the present time to make a statement about the impact of variations in scenario difficulty, concerning either formal or preformal students, owing to the varying difficulty of the five tests.

Our final hypothesis was that the use of material accessories to represent nonobservable phenomena would help facilitate an understanding of these phenomena. Our results show that this did indeed happen, but success is related more to learning style than to operational levels of reasoning. This last conclusion points to intriguing possibilities in the study of student motivation. Earlier research has pointed out the favorable impact of simulation games on motivation (among others, Bredemeier & Greenblat, 1981; Cherryholmes, 1966; Lee & O'Leary, 1971). Our results suggest that this may be due to the freedom of movement allowed by games. Those students whose styles involved movement and social interaction were those most favorable to the game. In attempting to use a simulation game to concretize abstract concepts, we are reaching out to those aspects of learning.

This brings us back to our opening remarks: Does teaching become abstract too soon? Perhaps we should say instead (or as well) that teaching becomes sedentary too soon, forgetting the oft-repeated injunctions of masters such as Froebel and Montessori (about the importance of movement) and Piaget (about the coordination of actions). We also noted a favorable reaction to the game of those students preferring more social styles of learning. In this light, it is worth noting that active involvement was the only affective factor significantly linked to learning. We might generalize this by saying that simulation games can help motivate social-minded students.

These two remarks should help guide further research on teaching with simulation games. We must try to give students mobile and tactile instruments, which they can manipulate themselves as tools to study and understand ideas and abstract concepts. Games must also allow participants to discuss among themselves hypotheses, methods, and lines of approach in terms of situation analysis and choice of strategy. A game with predetermined results and behavior is no longer, in our sense, a game.

Authors' Note

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Declaration of Conflicting Interests

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Bios

Pierre Corbeil holds degrees from the University of Toronto (Ontario, Canada) and the Université de Montréal (Québec, Canada). After focusing on war-gaming, historical games, and inter-cultural games, he became interested in entrepreneurship and subsequently published a collection of games on that subject, *Entreprendre par le jeu* (Becoming an entrepreneur through games). He has recently joined an international group called La Ludobanque (The Game Bank). He believes that games favor invention over classification, and he defines history as the illusion of reality re-created generation after generation. He still plays war games, in addition to writing science fiction (in French).
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